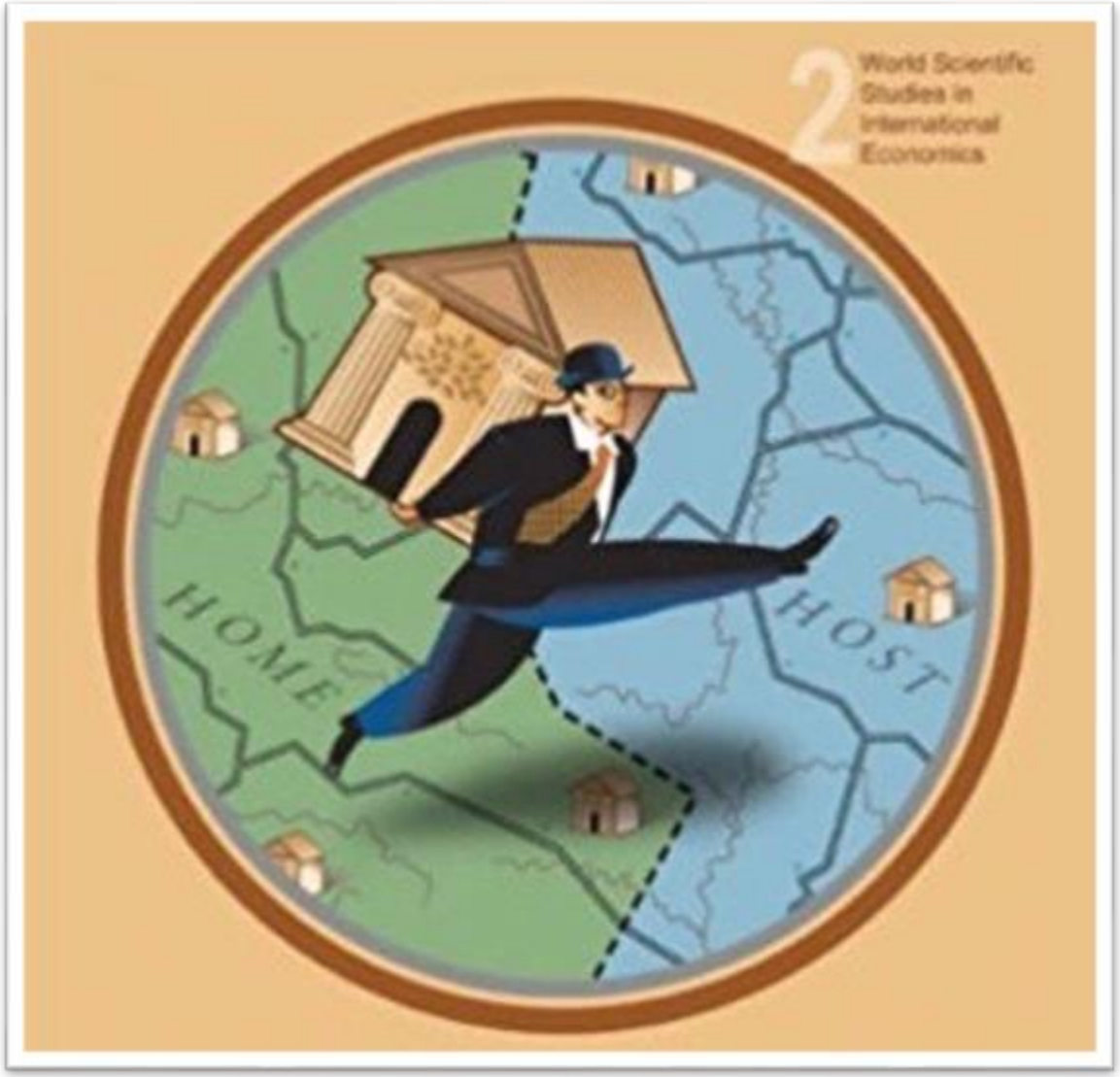


Lecture: International Finance (under flexible exchange rates)

Pay attention to the notes written under the slides

Explore Econ 2026
May 28th

Explore Econ 2026
May 28th

Celebrating 200 years of UCL Economics

Explore Econ 2026

- Our undergraduate research conference.
- A great opportunity for you to present your research to academics and economist from corporate/public sector.
- Excellent way to signal your research skills in your CV.
- There will be awards.
- There will be a judging panel.

Key dates for the 2026 competition

Monday 26 January	Paper and poster entries opens
Friday 20th March	Priority Submission deadline for submissions *Good quality papers submitted by this deadline will be given priority consideration*
Wednesday 29th April	Final deadline for any submissions
Monday 14 May	Peer voting for the People's Choice submission opens
Monday 21 May	Peer voting for the People's Choice submission closes
Thursday 28 May	Explore Econ conference, where the winning submissions are presented. Followed by an End of Year drinks and BBQ celebration, where we'll send the year off in style

Support your peers and fill this survey please.

Lecture plan:

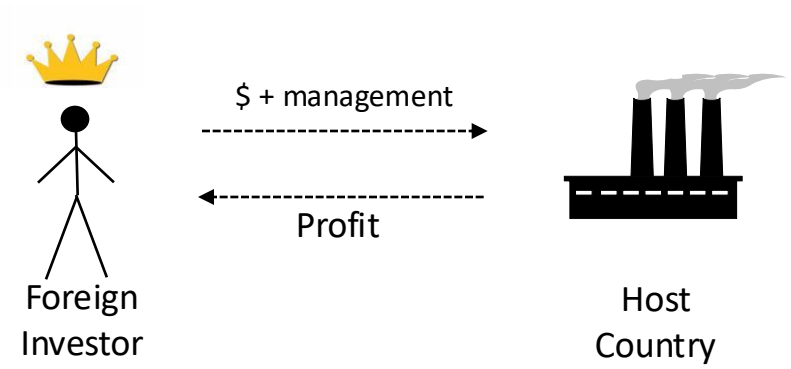
- Different types of foreign capital flows
- Link between international finance and international trade
- Floating nominal exchange rates
- The impact of trade flows on floating exchange rate
- The impact of capital flows on floating exchange rate

Lecture plan:

- **Different types of foreign capital flows**
- Link between international finance and international trade
- Floating nominal exchange rates
- The impact of trade flows on floating exchange rate
- The impact of capital flows on floating exchange rate

Foreign investments

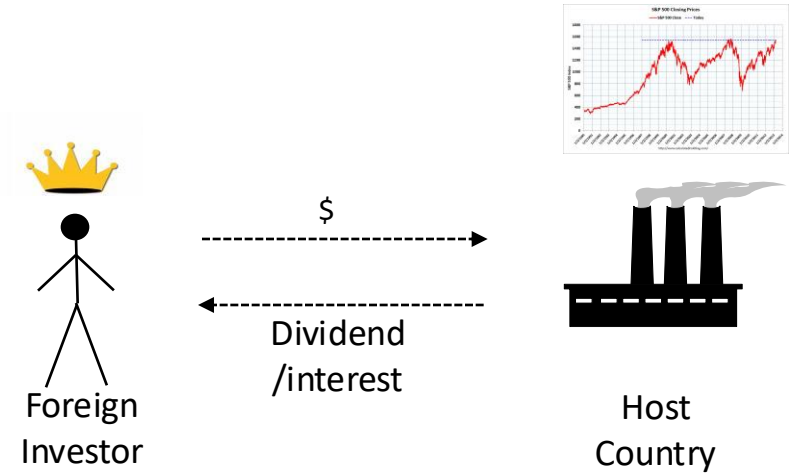
- Foreign direct investment



Foreign investments

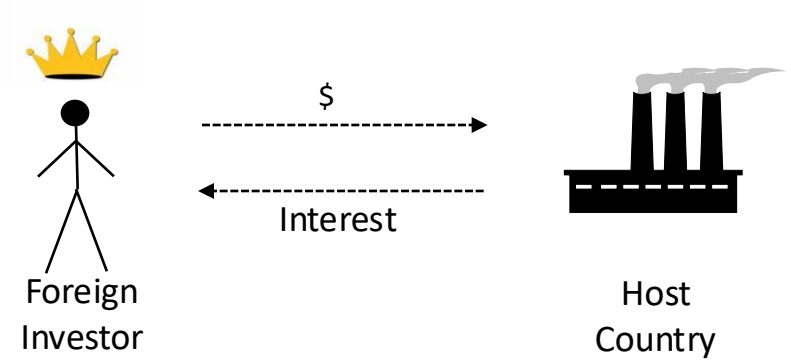
- Foreign direct investment

- Portfolio investment



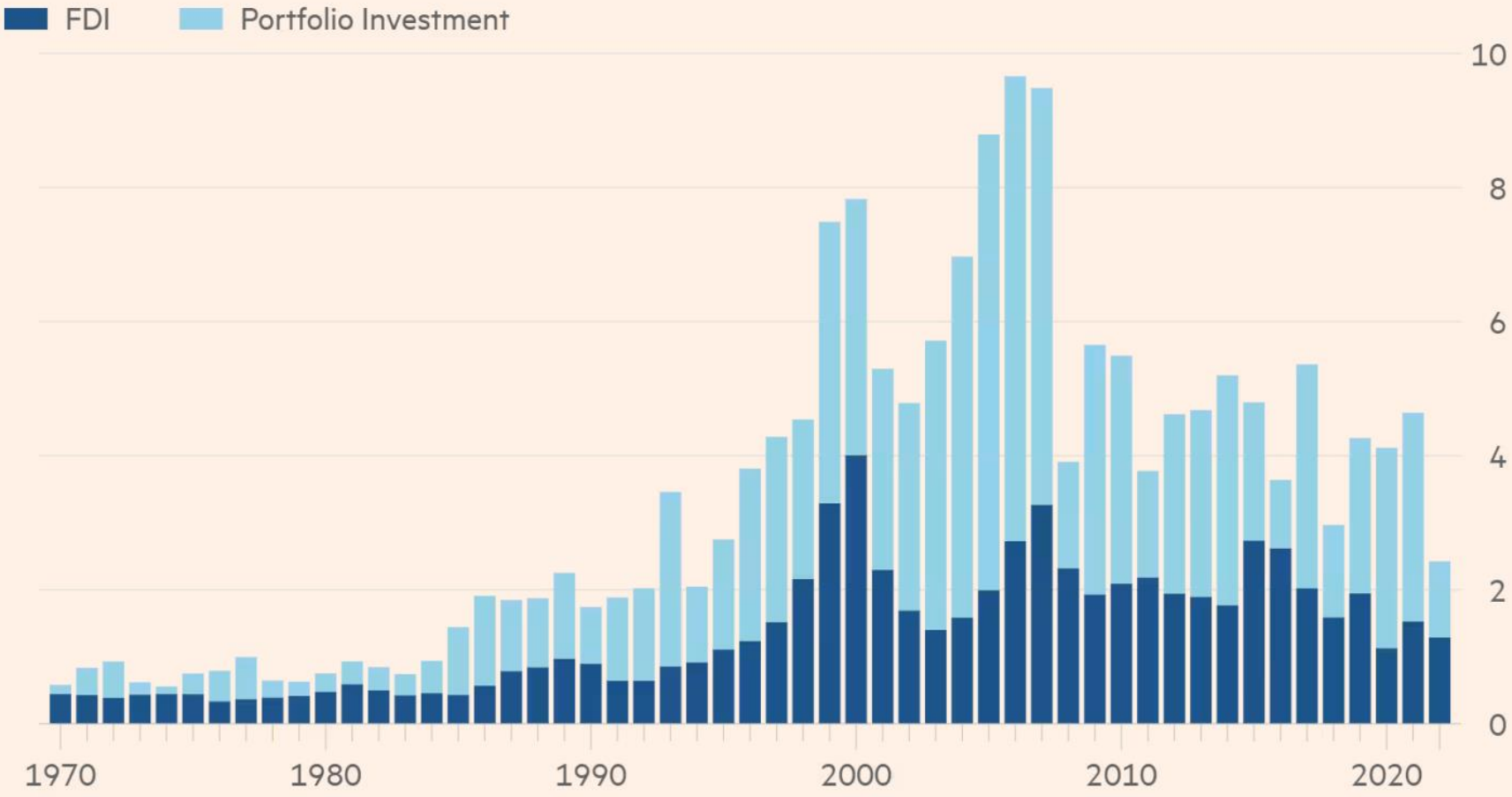
Foreign investments

- Foreign direct investment
- Portfolio investment
- Loan inflows



Financial globalisation peaked just before the 2008 financial crisis

Global gross financial flows (inward flows, as a % of global GDP)



Source: Subramanian, Kessler, & Properzi (Peterson Institute, 2023)

© FT

Lecture plan:

- Age of financial globalisation
- **Link between international finance and international trade**
- Floating nominal exchange rates
- The impact of trade flows on floating exchange rate
- The impact of capital flows on floating exchange rate

Small export business in China:

Small export business in China:

Current account: export-import of machinery & leather
($32,000-$20,000)

+$12,000

Small export business in China:

Current account: export-import of machinery & leather
($32,000-$20,000)

+$12,000

Capital account: net exporter of savings/capital to abroad

-$12,000

Small export business in China:

Current account: export-import of machinery & leather **+$12,000**

($32,000-$20,000)

Capital account: net exporter of savings/capital to abroad **-$12,000**

Lending dollars to a foreign bank
Buying shares of a foreign company

Small export business in China:

Current account: export-import of machinery & leather
($32,000-$20,000)

+$12,000

Small export business in China:

Current account: export-import of machinery & leather
($32,000-$20,000)

+$12,000

Changes in Central bank's foreign exchange reserves:

-$12,000

Small export business in China:

Current account: export-import of machinery & leather
($32,000-$20,000)

+$12,000

Small export business in China:

Current account: export-import of machinery & leather **+$12,000**

($32,000-$20,000)

Capital account: net exporter of savings/capital to abroad **-$4,000**

Lending dollars to a foreign bank
Buying shares of a foreign company

Changes in Central bank's foreign exchange holdings: **-$8,000**

Let’s aggregate to the country-level:

Current account: export-import of machinery & leather

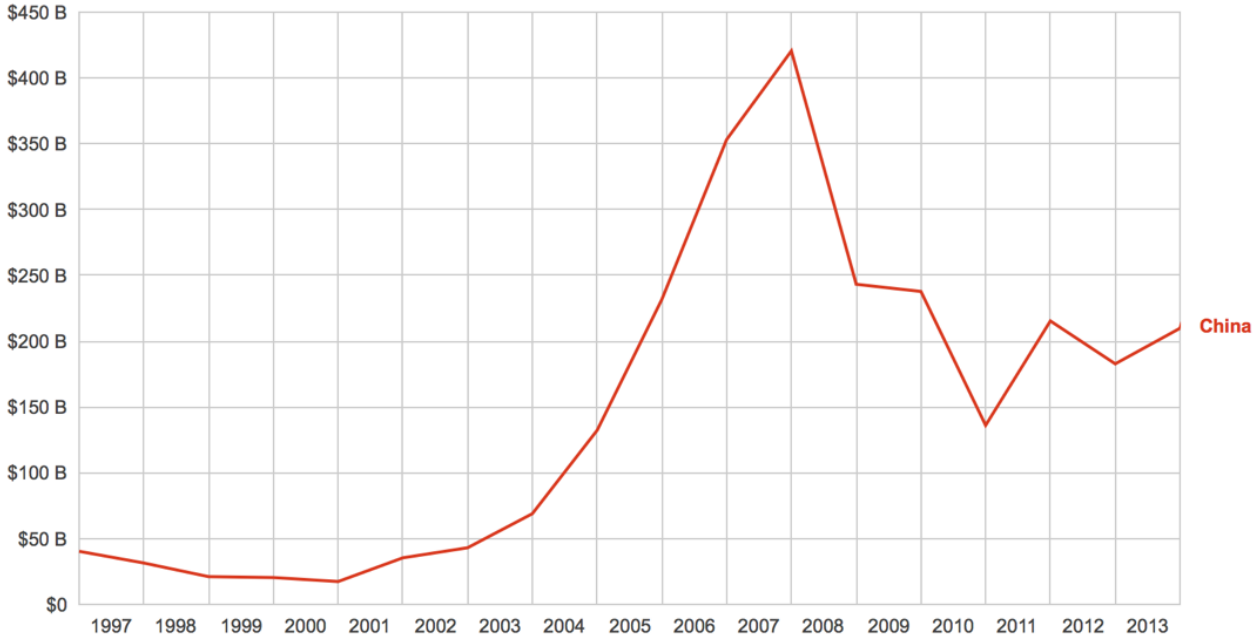
Capital account: Import of capital – export of capital

***Changes* in central bank’s foreign exchange reserves (Official reserve transactions)**

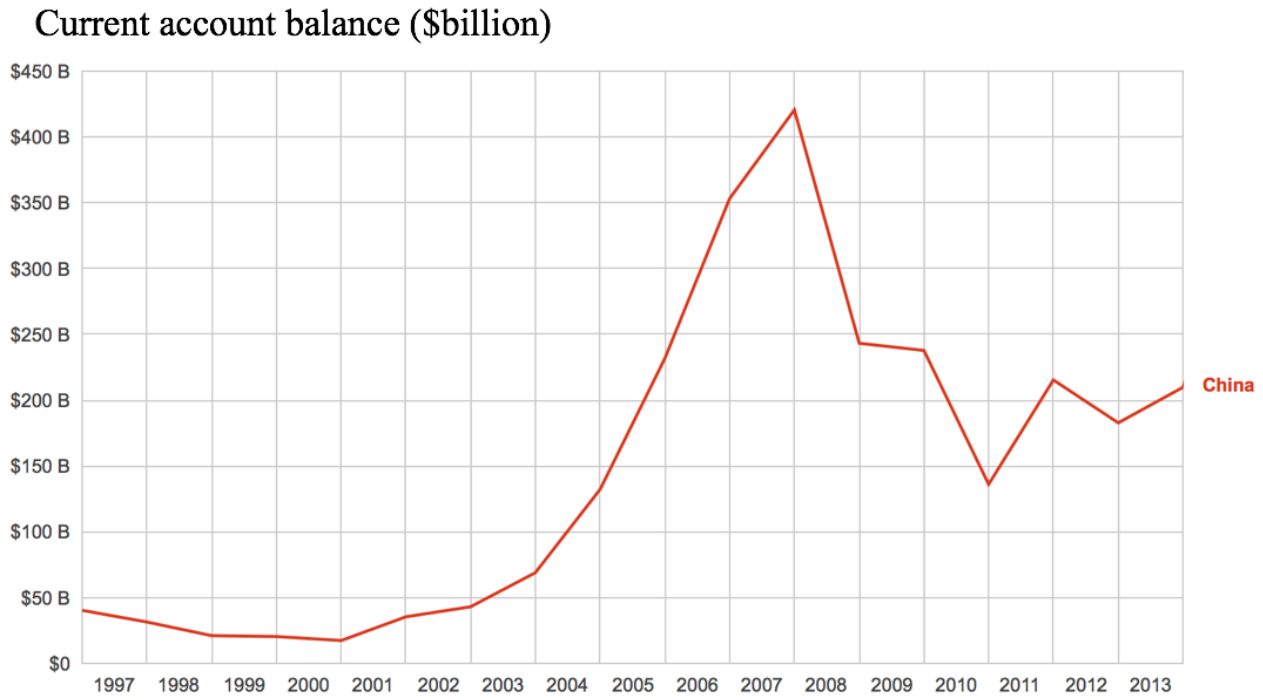
Country level:

$$\text{Current Account} + \text{Capital/Financial Account} + \text{Official Reserve Transactions} = 0 \tag{13.7}$$

Current account balance ($billion)



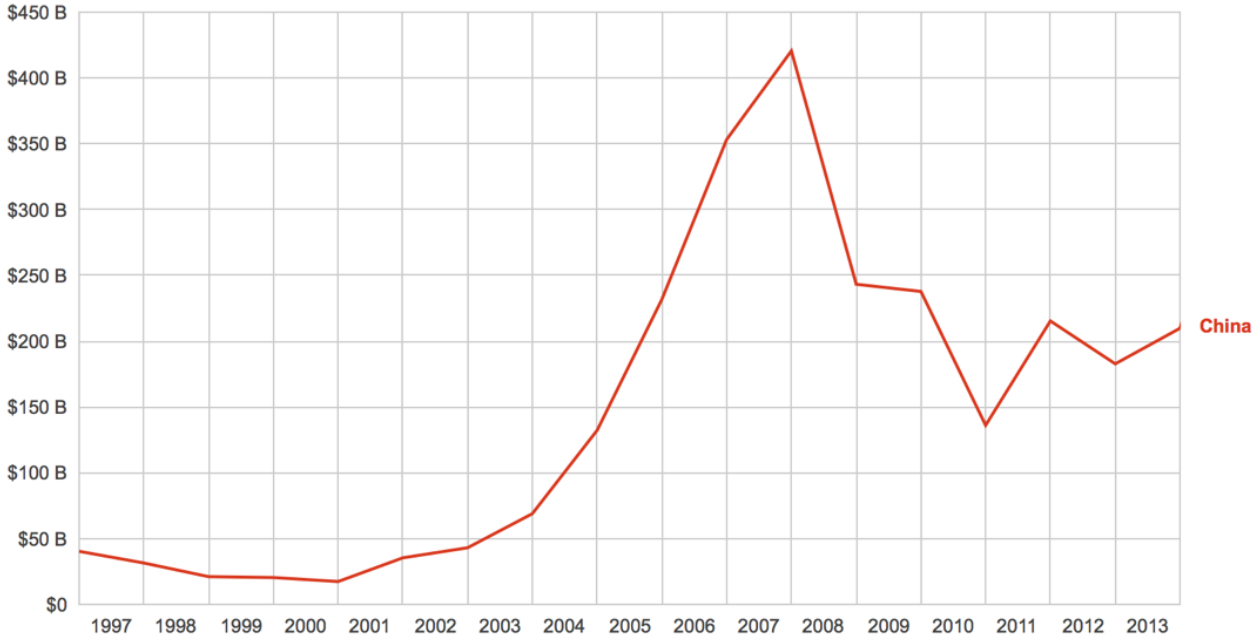
Source: Google Public Data (2017)



Source: Google Public Data (2017)

China's earnings (exports) has been higher than its spending (imports) for most of the time. This means Chinese firms has had a surplus of dollars.

Current account balance ($billion)



Source: Google Public Data (2017)

China's earnings (exports) has been higher than its spending (imports) for most of the time. This means Chinese firms has had a surplus of dollars.

Two things could have happened to this surplus:

1. Chinese firms has been exporting this capital abroad (capital account deficit)

2. Or Chinese firms has been selling their dollars to the Chinese central bank (reserve accumulation).

Small export business in Turkey:

Small export business in Turkey:

Current account: export-import of machinery
($4,000-$10,000)

-$6000

Small export business in Turkey:

Current account: export-import of machinery **-$6000**

($4,000-$10,000)

Capital account: net importer of savings/capital from abroad **+$6000**

Borrowing dollars from a foreign bank (selling IOU)
Selling shares to a foreign investor

Small export business in Turkey:

Current account: export-import of machinery
($4,000-$10,000)

-$6000

Changes in Central bank's foreign exchange holdings:

+$6000

Small export business in Turkey:

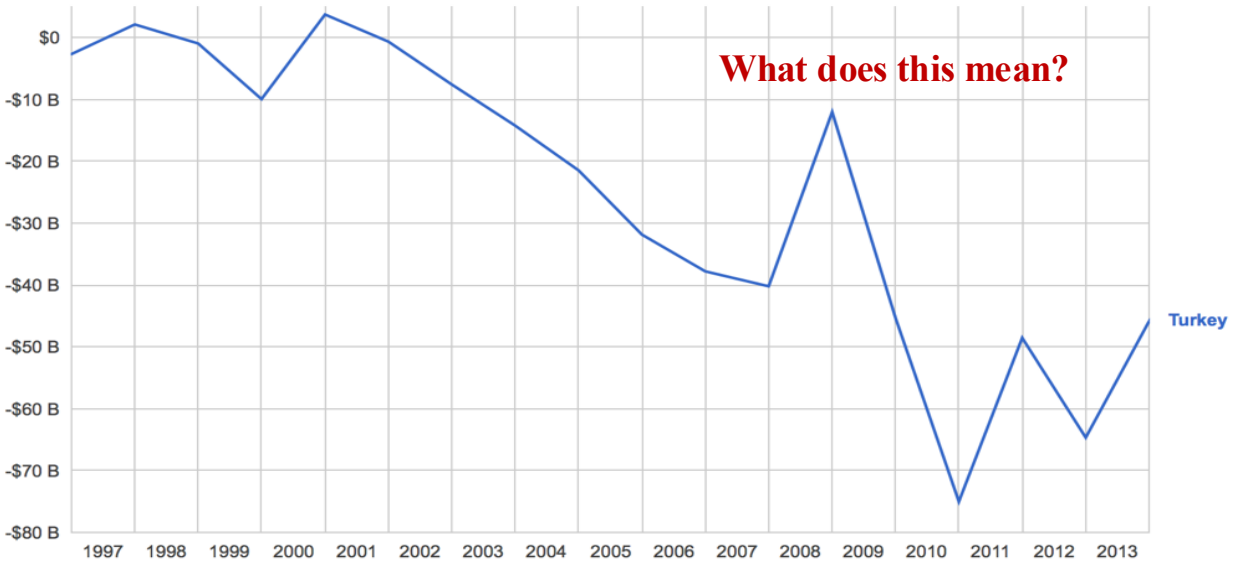
Current account: export-import of machinery
($4,000-$10,000) **-$6000**

Capital account: net importer of savings/capital from abroad **+$4000**

Borrowing dollars from a foreign bank (selling IOU)
Selling shares to a foreign investor

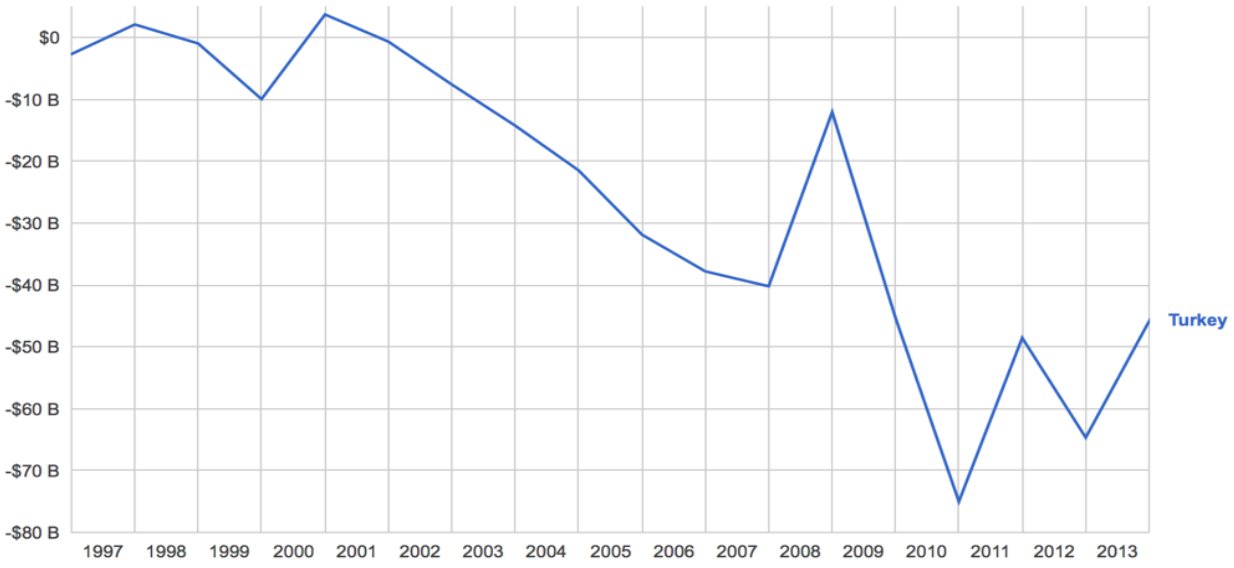
Changes in Central bank's foreign exchange holdings: **+$2000**

Current account balance ($billion)



Source: Google Public Data (2017)

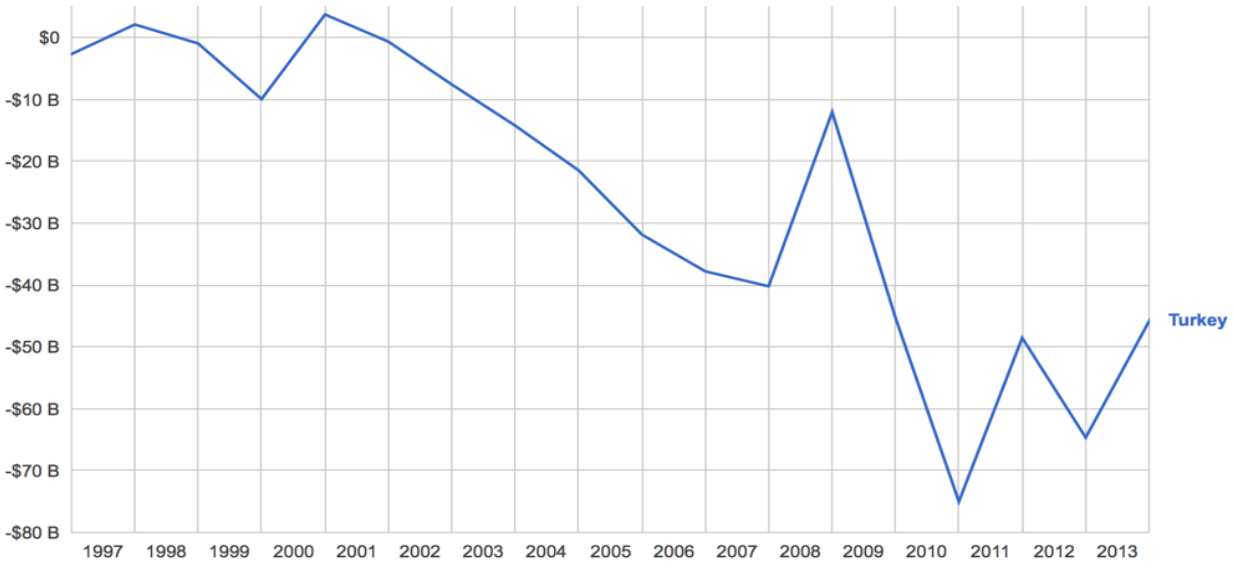
Current account balance ($billion)



Source: Google Public Data (2017)

Turkey's earnings (exports) has been lower than its spending (imports). So Turkish firms has had a deficit of dollars

Current account balance ($billion)



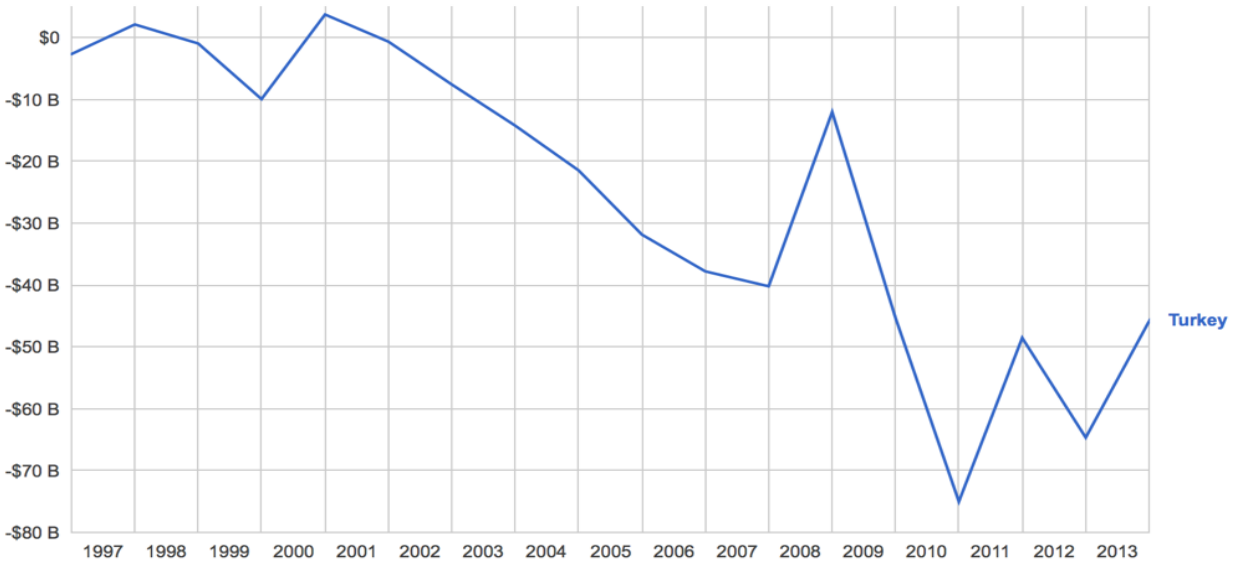
Source: Google Public Data (2017)

Turkey's earnings (exports) has been lower than its spending (imports). So Turkish firms has had a deficit of dollars

This deficit of dollars could have been financed in two ways:

1. Turkish firms have been importing capital (dollars) from abroad.

Current account balance ($billion)



Source: Google Public Data (2017)

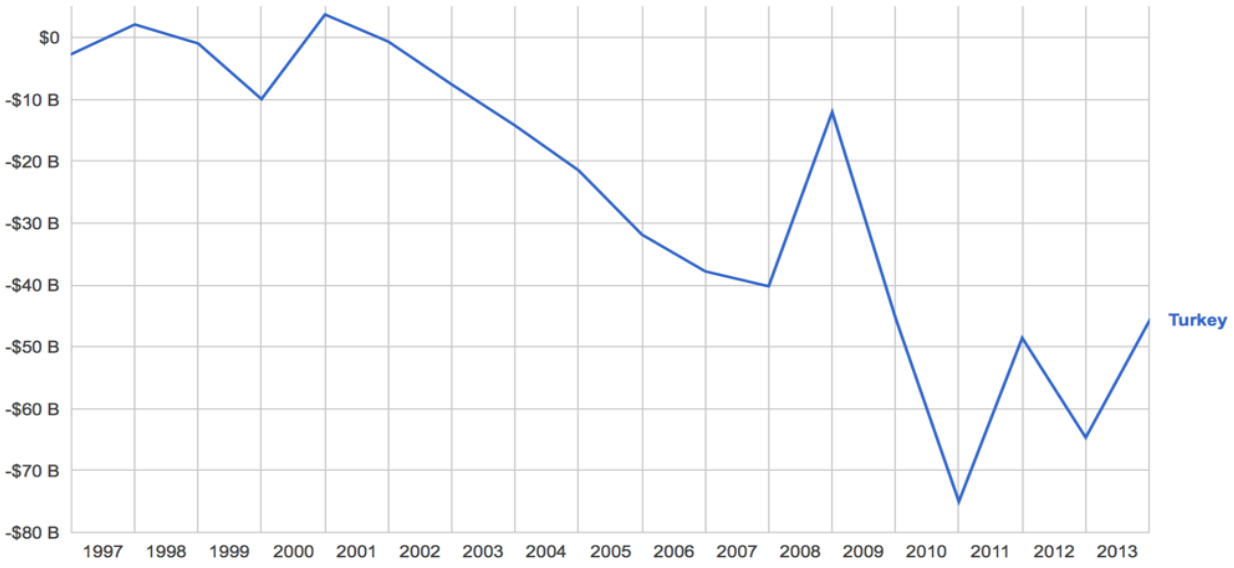
Turkey's earnings (exports) has been lower than its spending (imports). So Turkish firms has had a deficit of dollars

This deficit of dollars could have been financed in two ways:

1. Turkish firms have been importing capital (dollars) from abroad.

2. Turkish firms have been buying dollars from the Turkish central bank (running down the official reserves)

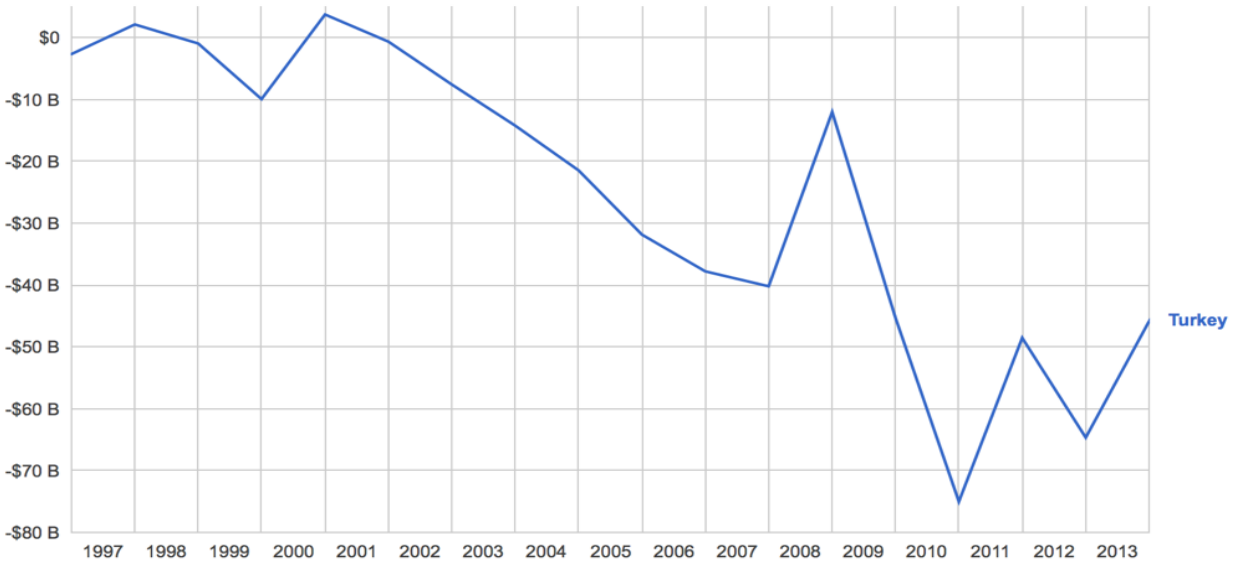
Current account balance ($billion)



Source: Google Public Data (2017)

Why running a persistent and growing current account deficit is bad?

Current account balance ($billion)



Source: Google Public Data (2017)

Why running a persistent and growing current account deficit is bad?

To finance a growing current account, countries can do two things.

1. Run down their foreign exchange reserves (their past dollar savings), which is not sustainable in the long-term. Eventually the country runs out of foreign exchange and has to do a painful adjustment: drastically reduce its imports and increase its exports.

2. The country has to heavily rely on foreign inflow of capital. But this makes your economy very vulnerable; if these inflows suddenly stop; you are forced to go to sudden painful adjustments (cut down your imports and increase your exports).

INSIDE: AN 18-PAGE SPECIAL REPORT ON THE FUTURE OF BANKING
The Economist
MAY 19TH–25TH 2012
Economist.com
The decline of the public company
California's big prison reform
Egypt's promising election
What's going wrong with Brazil
Turkey's culture war
The Greek run
Y20754779232
EVRO 20

Current account balance (% of GDP)

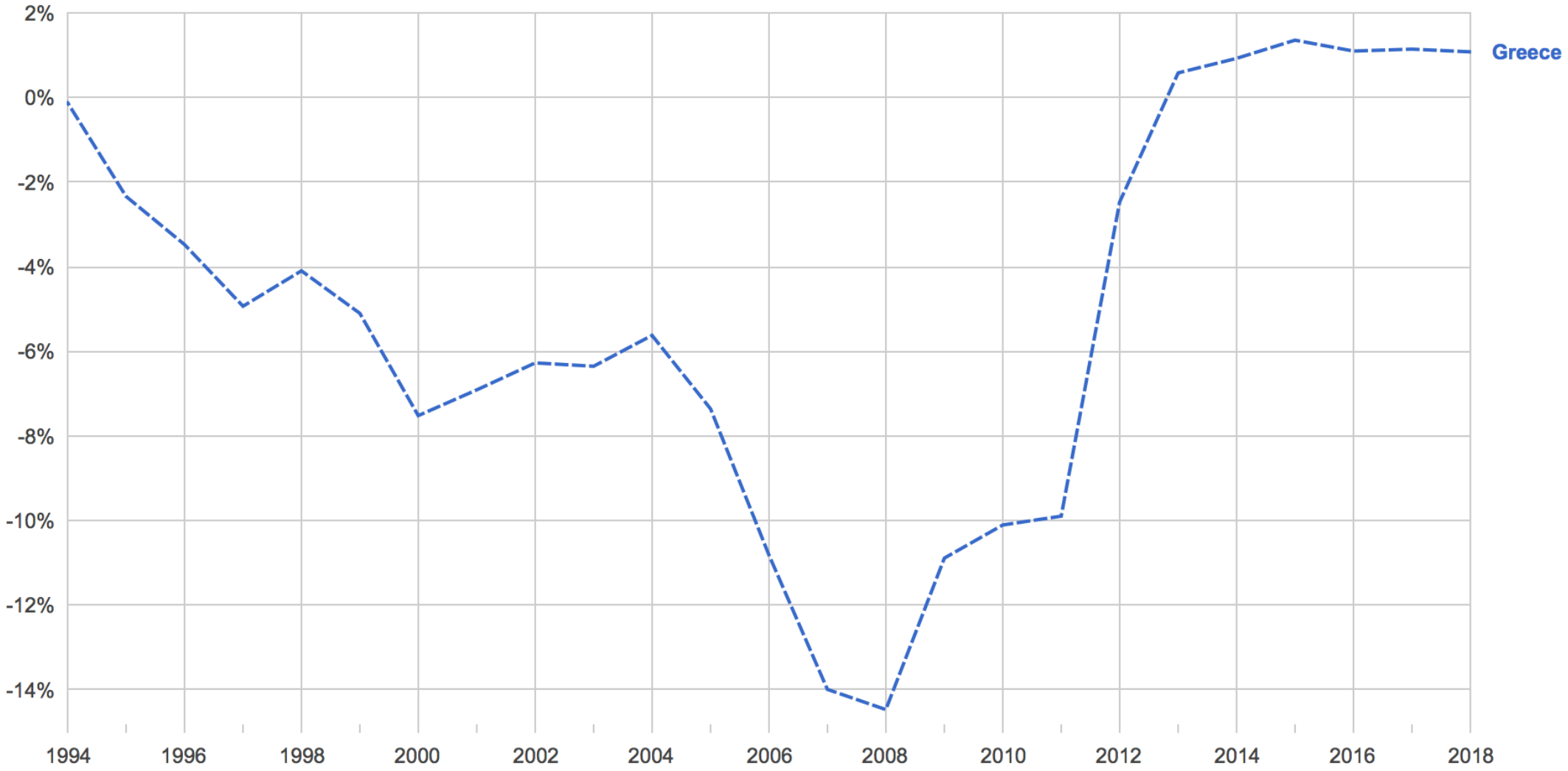
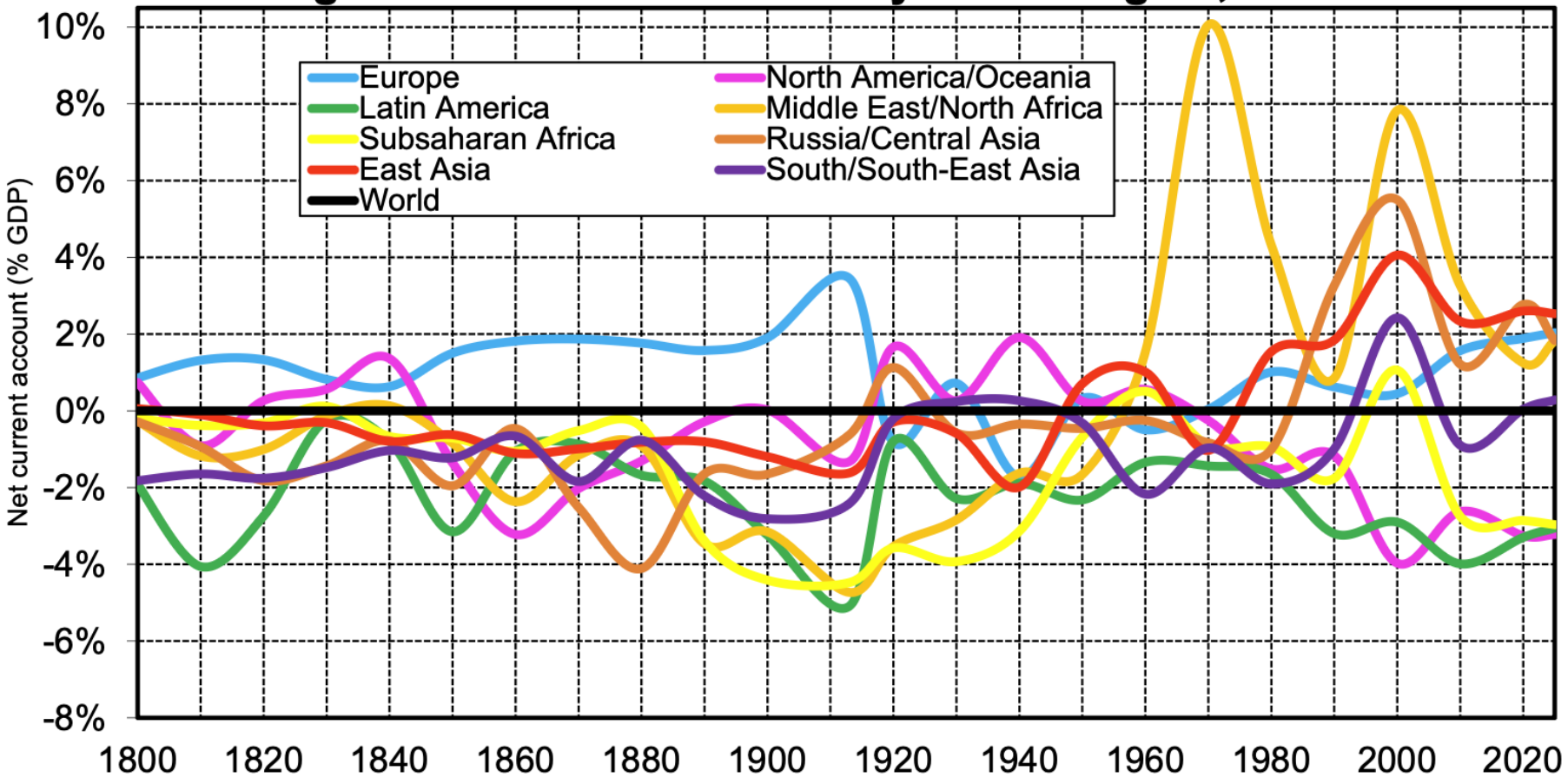
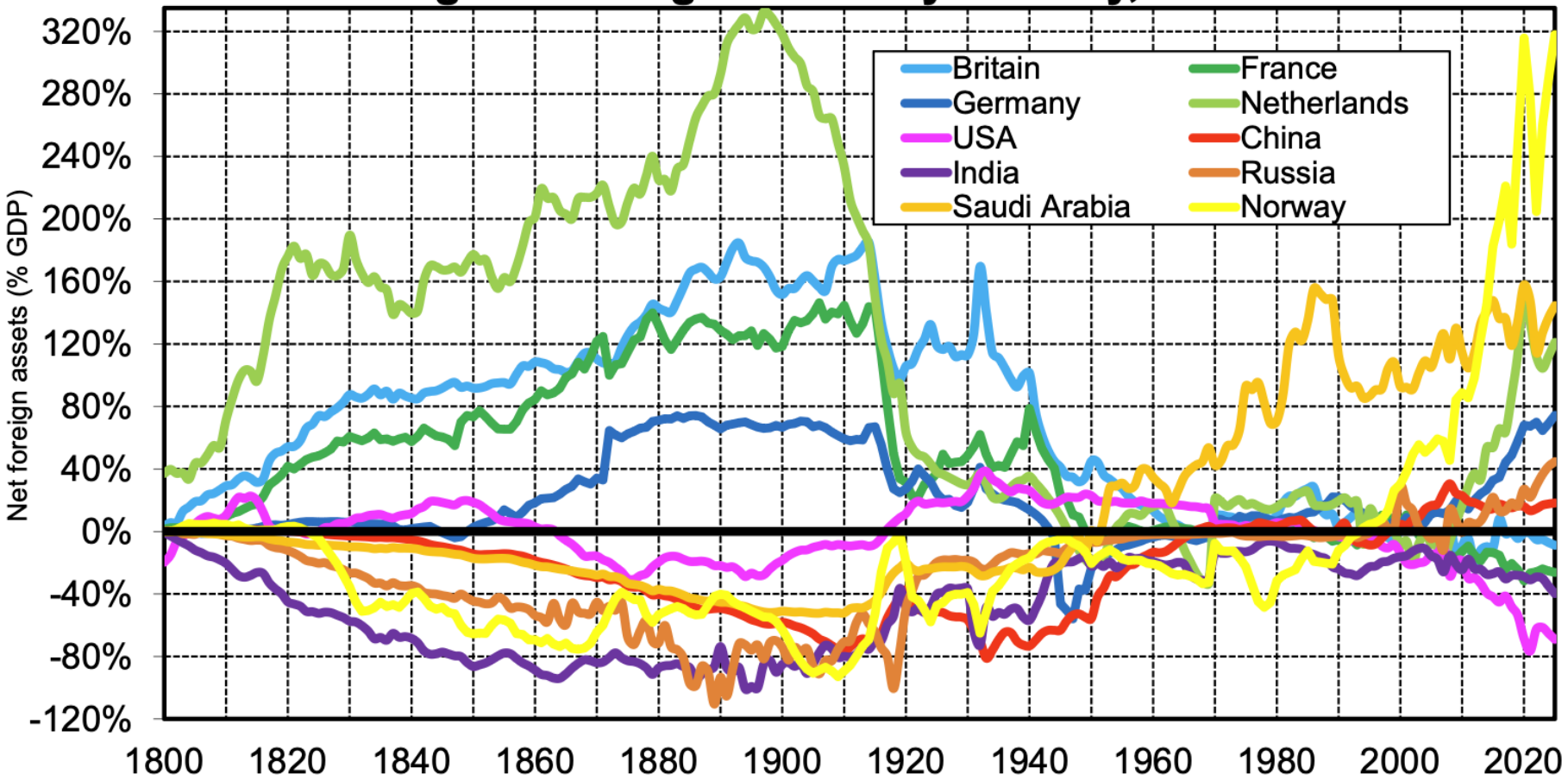


Fig. 8. Net Current Account by World Region, 1800-2025



Interpretation. Between 1800 & 1914, Europe has a permanent current account surplus (close to 2% of its GDP on average, and rising over time) while the rest of the world has a permanent deficit. Since the 1970s-1980s, the main surpluses come from oil countries (Middle East, Russia) and East Asia. **Note**. The values reported here are decennial averages: 1800 refers to 1800-1809, 1810 to 1810-1819, etc. **Sources and series**: see wid.world

Fig. 14. Foreign Wealth by Country, 1800-2025



Interpretation. If we include smaller economies into the picture, we find that net foreign assets can be as large as 300% of a country's GDP or more, such as the Netherlands in 1900 (a small country with large colonial holdings in Indonesia) or Norway in 2025 (a small country with enormous oil and gas reserves that were transformed into a large sovereign fund in a recent decades).
Sources and series: wid.world

Lecture plan:

- Different types of foreign capital flows
- Accounting link between international finance and international trade
- **Floating nominal exchange rates**
- The impact of trade flows on floating exchange rate
- The impact of capital flows on floating exchange rate

What are the upsides and the downsides of having a floating exchange rate?

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

1$= 12 pesos

Supply of dollars
Demand for pesos

Demand for dollars
Supply of pesos

1$= 12 pesos

Supply of dollars
Demand for pesos

Depreciation of Pesos

1$= 16 pesos

Demand for dollars
Supply of pesos

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

1$= 12 pesos

<u>Supply of dollars</u>
Demand for pesos

<u>Demand for dollars</u>
Supply of pesos

1$= 12 pesos

Supply of dollars
Demand for pesos

Appreciation of Pesos

1$= 8 pesos

Demand for dollars
Supply of pesos

1$=12 Mexican pesos

Nominal exchange rate= 12 pesos/dollar

$$e = \frac{pesos}{dollar}$$

$$e = \frac{home\ currency}{foreign\ currency}$$

$e \uparrow \Rightarrow$ value of the home currency (peso) *falls* or *depreciates*	Promotes Mexican exports
$e \downarrow \Rightarrow$ value of the home currency (peso) *rises* or *appreciates*	Promotes Mexican imports

$$\text{Real exchange rate for Mexico} = \frac{\text{Domestic prices in US (\$)}}{\text{Domestic prices in Mexico (pesos)}} \times e = \frac{\text{Domestic prices in US (Pesos)}}{\text{Domestic prices in Mexico (Pesos)}}$$

An <u>increase in *e* (depreciation of pesos)</u>= increases the relative price of American goods to Mexican goods. Hence <u>good for Mexican exports</u> and bad for US exports.

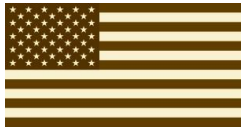
A <u>decrease in *e* (appreciation of pesos)</u>= decreases the relative price of American goods to Mexican goods. Hence <u>bad for Mexican exports</u> and good for US exports.

Lecture plan:

- Different types of foreign capital flows
- Accounting link between international finance and international trade
- Floating nominal exchange rates
- **The impact of trade flows on floating exchange rate**
- The impact of capital flows on floating exchange rate

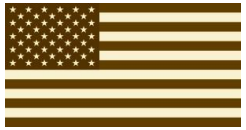
A trade-based model of floating exchange rates:

Inflation differences across countries $\xrightarrow{?}$ Trade flows



exports

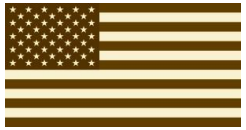
imports



exports

imports

INTERNATIONAL CURRENCY EXCHANGE



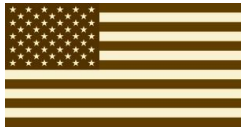
exports

$5 billion

(60 billion pesos)

imports

INTERNATIONAL CURRENCY EXCHANGE



exports

$5 billion

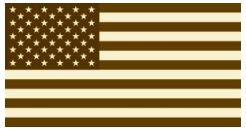
(60 billion pesos)

imports

INTERNATIONAL CURRENCY EXCHANGE

imports

exports



exports

$5 billion

(60 billion pesos)

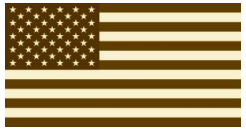
imports

imports

$5 billion

(60 billion pesos)

exports



exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

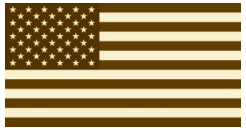
ice INTERNATIONAL CURRENCY EXCHANGE

imports

$5 billion

(60 billion pesos)

exports



exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

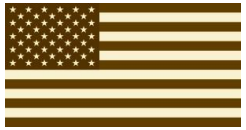
10%
inflation

imports

$5 billion

(60 billion pesos)

exports



What will happen to Mexico's trade balance?

exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE

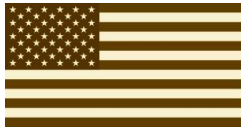
10% inflation

imports

$5 billion

(60 billion pesos)

exports



What will happen to Mexico's trade balance?

exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

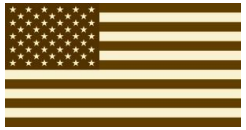
10%
inflation

imports

$5 billion

(60 billion pesos)

exports



What will happen to Mexico's trade balance?

exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

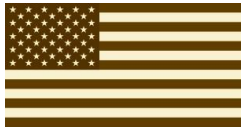
10%
inflation

imports

$4 billion

(48 billion pesos)

exports



Will Mexico's trade deficit persist?

exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE

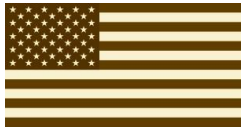
10% inflation

imports

$4 billion

(48 billion pesos)

exports



Will Mexico's trade deficit persist? NO!

exports

$5 billion

(60 billion pesos)

imports

1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE

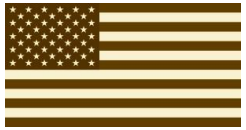
10% inflation

imports

$4 billion

(48 billion pesos)

exports



Will Mexico's trade deficit persist? NO!

exports

$5 billion

(60 billion pesos)

imports

Depreciation 10%

1$= 13.2 pesos

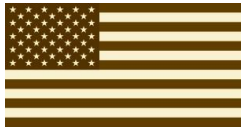
10% inflation

imports

$4 billion

(48 billion pesos)

exports



Will Mexico's trade deficit persist? NO!

exports

$5 billion

(60 billion pesos)

imports

Depreciation 10%

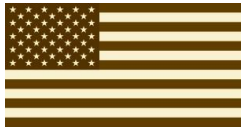
1$= 13.2 pesos

10%
inflation

imports

$5 billion
(66 billion pesos)

exports



Will Mexico's trade deficit persist? NO!

In long-term, nominal currency depreciates and neutralises the effect of inflation

exports

$5 billion

(60 billion pesos)

imports

Depreciation 10%

1$= 13.2 pesos

INTERNATIONAL CURRENCY EXCHANGE

10% inflation

imports

$5 billion

(66 billion pesos)

exports

Currency depreciation
in the long term

10%

constant

$$\frac{\text{Domestic prices in US (\$)}}{\text{Domestic prices in Mexico (pesos)}} \times e = \frac{\text{Domestic prices in US (Pesos)}}{\text{Domestic prices in Mexico (Pesos)}}$$

10%
Domestic inflation

Trade-based model of floating exchange rates:

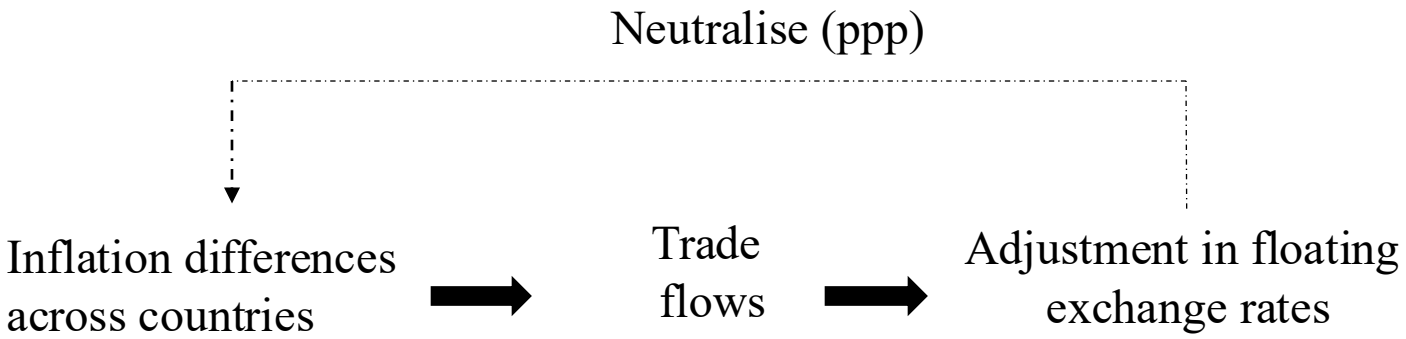
Trade-based model of floating exchange rates:

Inflation differences across countries → Trade flows

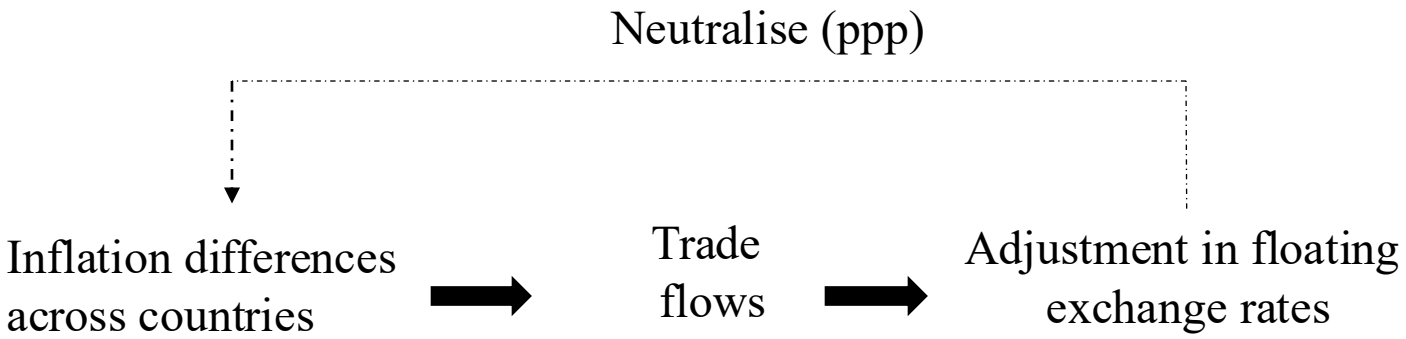
Trade-based model of floating exchange rates:

Inflation differences across countries → Trade flows → Adjustment in floating exchange rates

Trade-based model of floating exchange rates:



Trade-based model of floating exchange rates:



Nominal exchnage rate adjust until it neutralises the effect of inflation differences across countries & balance the trade account in the long-term.

Currency depreciation
in the long term

10%

constant

$$\frac{\text{Domestic prices in US (\$)}}{\text{Domestic prices in Mexico (pesos)}} \times e = \frac{\text{Domestic prices in US (Pesos)}}{\text{Domestic prices in Mexico (Pesos)}} = 1$$

10%
Domestic inflation

Currency depreciation
in the long term

10%

constant

$$\frac{\text{Domestic prices in US (\$)}}{\text{Domestic prices in Mexico (pesos)}} \times e = \frac{\text{Domestic prices in US (Pesos)}}{\text{Domestic prices in Mexico (Pesos)}} = 1$$

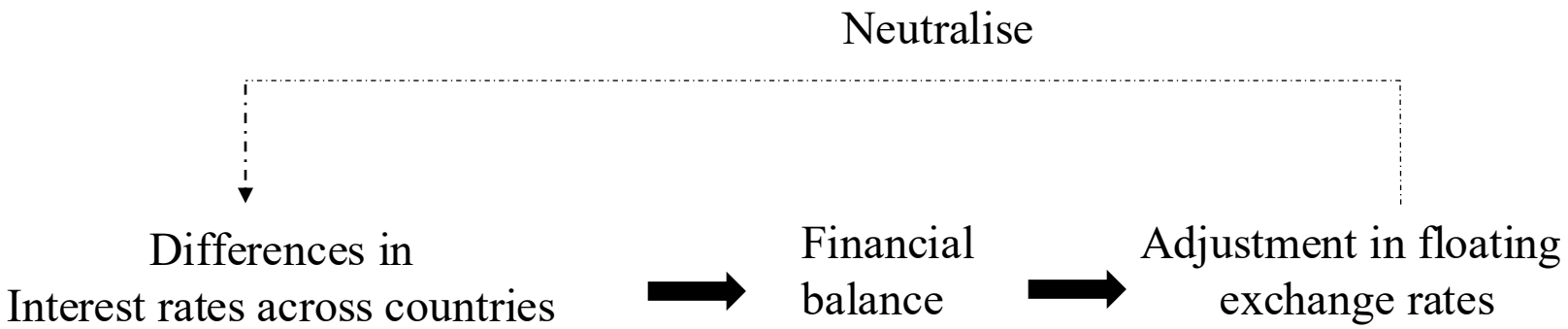
10%
Domestic inflation

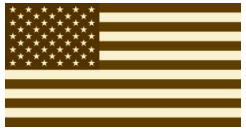
Purchasing power parity (PPP) in <u>the long-term</u>. Meaning that the ratio of price of goods in the US and Mexico remains the same. Hence, competitiveness of Mexico remains intact.

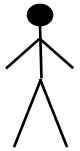
Lecture plan:

- Age of financial globalisation
- Accounting link between international finance and international trade
- Floating nominal exchange rates
- The impact of trade flows on floating exchange rate
- The impact of capital flows on floating exchange rate

Asset-based model of floating exchange rates:

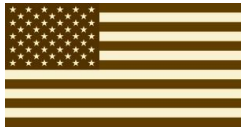


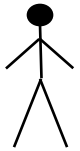




Investor

2%

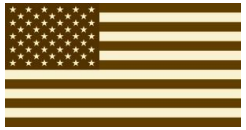


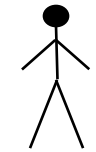


Investor

2%

1$= 12 pesos





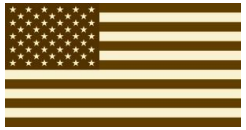
Investor

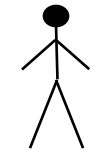
$5 billion
(60 billion pesos)

2%

1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE





Investor

$5 billion
(60 billion pesos)

2%

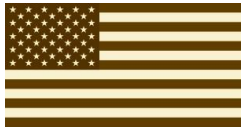
1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE

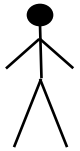
2%

$5 billion
(60 billion pesos)

Investor



Now assume interest rates in the US goes up to 5%. What will happen to Mexico's capital account balance?



Investor

$5 billion
(60 billion pesos)

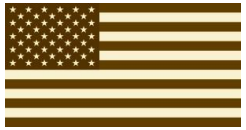
2%

1$= 12 pesos

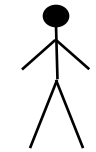
%5

$5 billion
(60 billion pesos)

Investor



Now assume interest rates in the US goes up to 5%. What will happen to Mexico's capital account balance?



Investor

$5 billion
(60 billion pesos)

2%

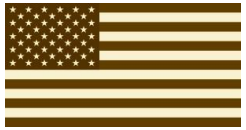
1$= 12 pesos

ice INTERNATIONAL CURRENCY EXCHANGE

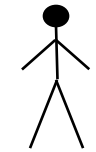
%5

$7 billion
(84 billion pesos)

Investor



Now assume interest rates in the US goes up to 5%. What will happen to Mexico's capital account balance?



Investor

$5 billion
(60 billion pesos)

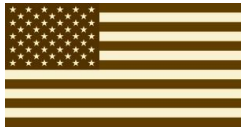
2%

1$= 12 pesos

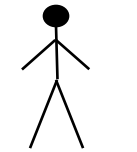
%5

$7 billion
(84 billion pesos)

Investor



Now assume interest rates in the US goes up to 5%. What will happen to Mexico's capital account balance?



Investor

$5 billion
(60 billion pesos)

2%

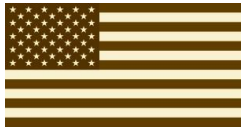
Depreciation 3%

1$= 12.36 pesos

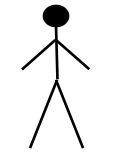
%5

$7 billion
(84 billion pesos)

Investor



Now assume interest rates in the US goes up to 5%. What will happen to Mexico's capital account balance?



Investor

$5 billion
(60 billion pesos)

2%

Depreciation 3%

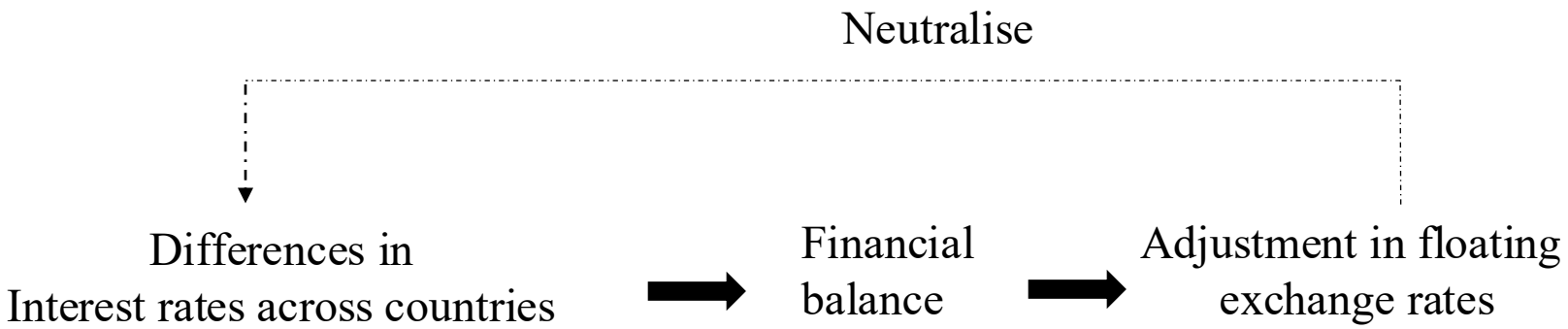
1$= 12.36 pesos

%5

$5 billion
(61.8 billion pesos)

Investor

Asset-based model of floating exchange rates:



If you have an interest rate differential across countries, then nominal exchange rate adjust until it neutralises the effect of that difference.

Power of the Market - Balancing Trade

LibertyPen

Subscribe 94K

13,802 views

Download Add to Share More 132 7

CLICK HERE

Summary

- International finance (exchange of assets) is closely aligned with international trade (exchange of goods). Dangerous of running a *large persistent* current account deficit.
- Flexible exchange rates; nominal value of currency is determined by demand & supply.
- Flexible exchange rate ensures that current account will be in balance in the long-term by adjusting to inflation differences across countries.
- Flexible exchange rate ensures capital account will be in balance in the long-term by adjusting to interest rate differentials across countries.